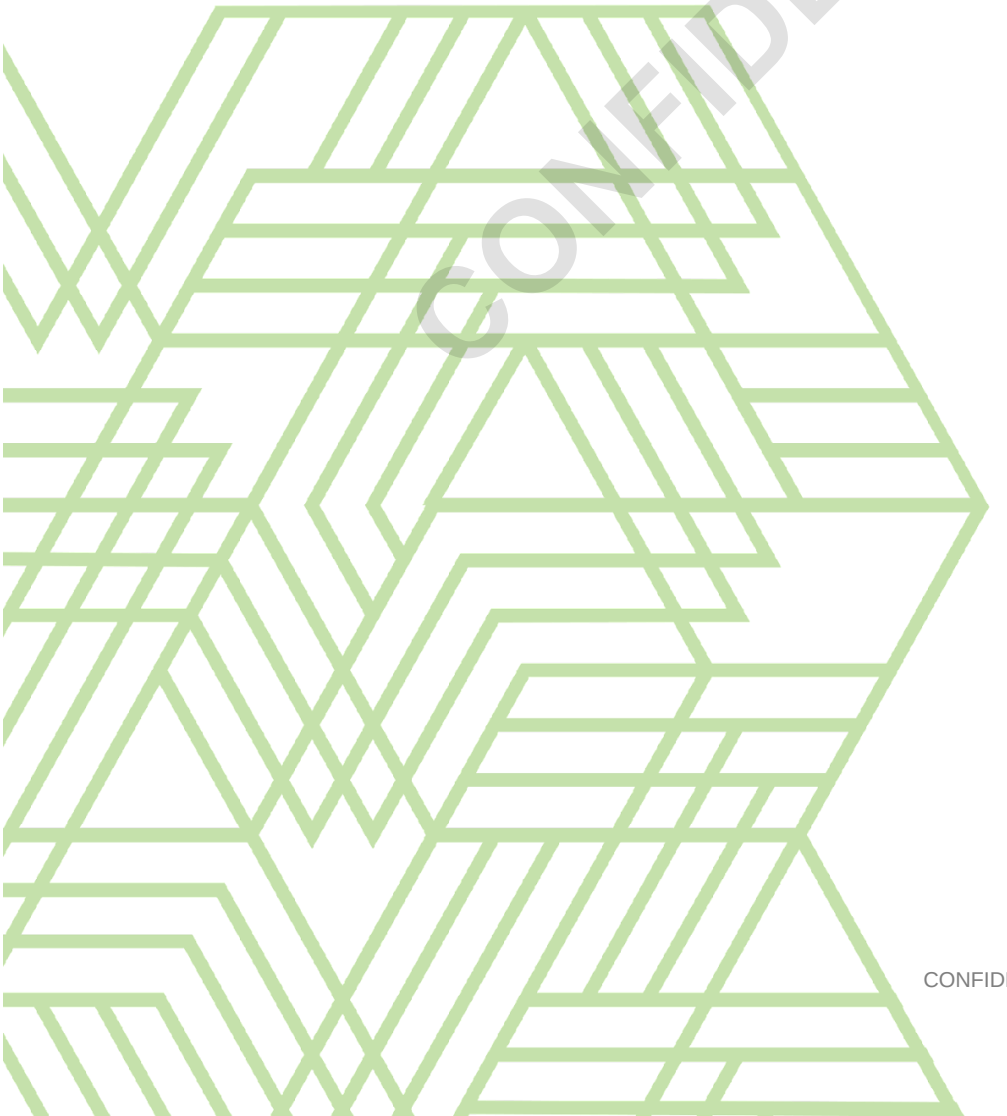
ATM33/e Bluetooth® Angle-of-Arrival (AoA) Demo

Quick Start Guide

SUMMARY: This document provides instructions on how to operate the Bluetooth Angle of Arrival demo with the ATM33/e wireless Bluetooth SoC.

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1. Overview

1.1 Angle of Arrival: A Brief Overview

Angle of Arrival (AoA) is a method of calculating the signal direction of a transmitting device, such as an asset tag that uses a single antenna to transmit a special signal. The tag transmits signals to a locator, typically stationary that has multiple antennas. Using phase differences of signals, the locator calculates angle and estimates the direction of the tag relative to itself.

Applications can include:

- Asset tracking
- Personal trackers
- Access control

1.2 AoA Using ATM33/e

Bluetooth's AoA technology that can support high-accuracy indoor location services and the ATM33/e (a.k.a ATM33/ATM33e) AoA evaluation kit consists of two (2) key hardware blocks which are preloaded with firmware. A Windows® GUI application (Graphical User Interface) package is available for download via the Atmosic Support website.

1.2.1 Locator

- 1x Antenna Array Board
- 1x ATM33/e evaluation board (preloaded with 'locator' firmware)
- 1x USB cable

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Figure 1 - Locator Antenna Array (Upper side)

The locator section (upper side) of the evaluation kit consists of an antenna array with the antennas configured in an inverted ‘L’ configuration.

In the center of the antenna board is an LED array which will be used to indicate relative positioning of the tag.

Three (3) additional LEDs in the top right, provide additional status information.



Figure 2 - Locator Antenna Array (Lower side)

The locator section (lower side) of the evaluation kit consists of the antenna switching circuit and a standard ATM33/e evaluation kit preloaded with firmware for locator functionality.
The antenna array is mounted to the ATM33/e evaluation kit via the 50-pin I/O header and a short coaxial cable. The 50-pin header provides the I/O connection for the LED display and antenna switching circuit. The locator is powered via a USB cable (provided).

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1.2.2 Tag

- 1x ATM33/e evaluation board (preloaded with ‘tag’ firmware)
- 1x 2.4 GHz Antenna
- 1x CR-2032 Coin battery

The tag capability is provided via a second ATM33/e evaluation board preloaded with tag firmware. The tag is powered via a supplied coin cell battery and transmits its beacon from the 2.4 GHz Bluetooth antenna which should be connected to the RFIO port.

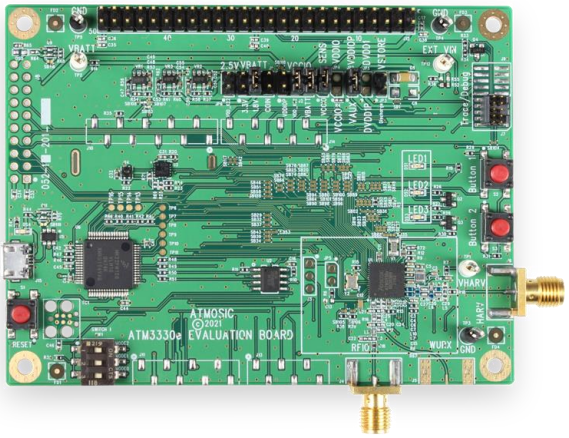


Figure 3 - ATM33/e Evaluation Board Preloaded with Tag Firmware

2. Using the AoA Demonstration Kit

The kit has been created using standard evaluation kits so developers can create their application code and perform any lab testing that they may choose after their initial evaluation.

This document assumes that developers have not changed any preloaded firmware but if they have, the firmware is also available as part of the SDK package.

2.1 Preparation Steps

- Verify the kit contents
- Install the GUI Software (Atmosic Utilities)
- Set up the ATM33/e EVB as an AoA Tag
- Set up the AoA Locator Board
- Launch GUI
- Using the demo

3. Installing the GUI Software

1) Do not plug any hardware into the Windows PC at this point

2) Download the Atmosic Utilities Zip File from the Atmosic website
   - Atmosic Utilities

3) After download is complete, unzip the compressed ‘Atmosic Utilities’ file.

4) From the unzipped “Atmosic Utilities” folder

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- Run ‘VC_redist.x64’ and follow instructions for installation.
- Open the embedded folder, v#.#.# where the # corresponds to the current version number of the software
- Open the x64 folder and then run the ‘Atmosic.Sales Tool.App’ file which has the msix extension to start the installer application

5) Ensure that ‘Launch when ready’ is checked as shown in Figure 4 below, as this will trigger the application to automatically start once the installation is complete.

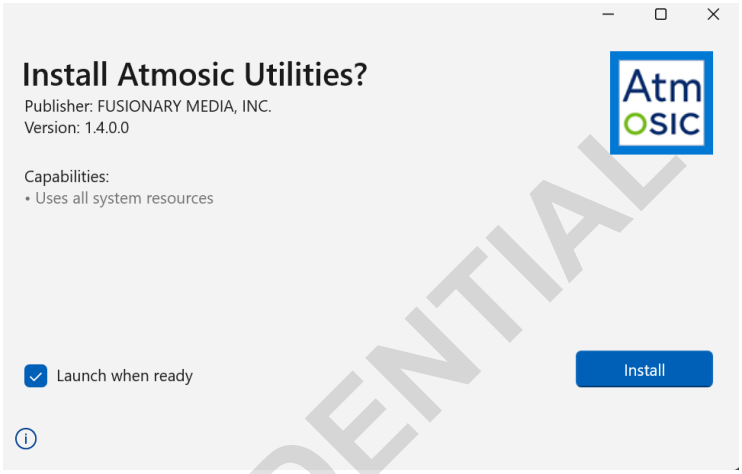


Figure 4 - Install Atmosic Utilities

6) After the GUI launches, it will try to communicate with the locator which has not been plugged in yet resulting in an error message, see Figure 5.

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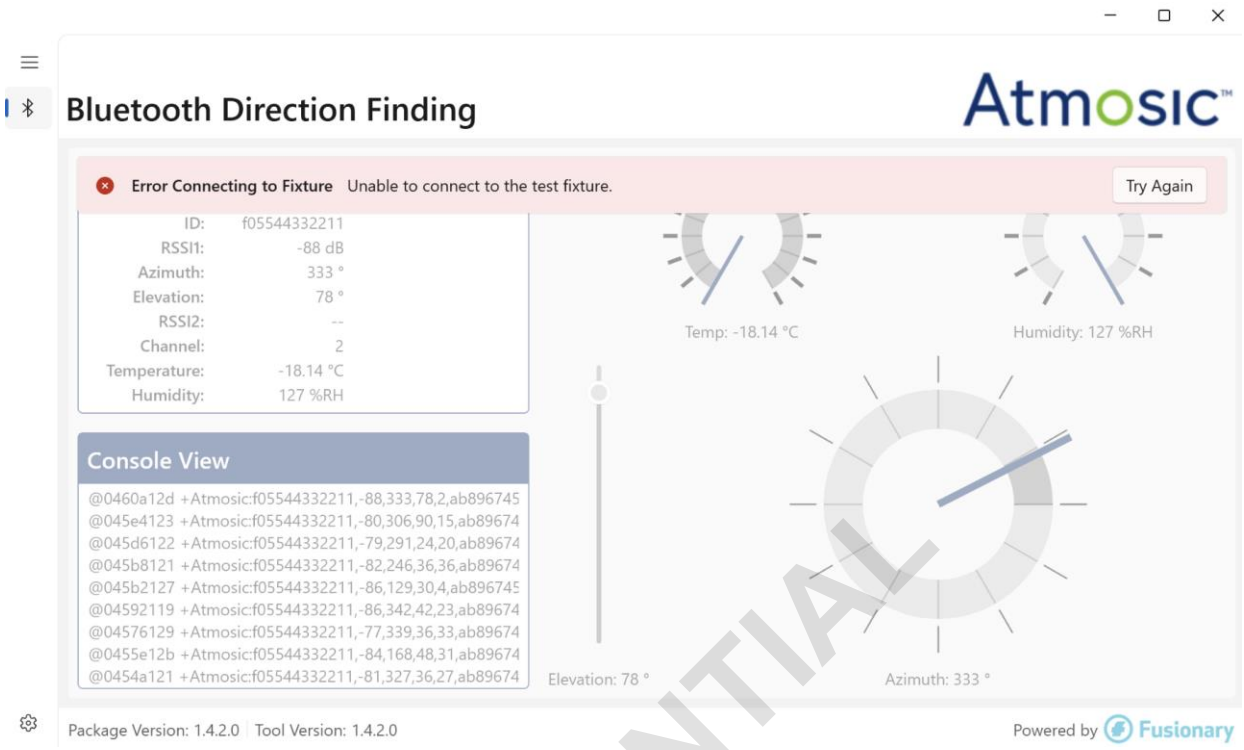


Figure 5 - Error Connecting to Fixture Message

***Note**: The interface of the GUI will be translucent because the locator has not been connected yet.*

4. Set Up the ATM33/e EVB As An AoA Tag

Connect the 2.4 GHz antenna to the RFIO connector of the EVB as shown in Figure 6.

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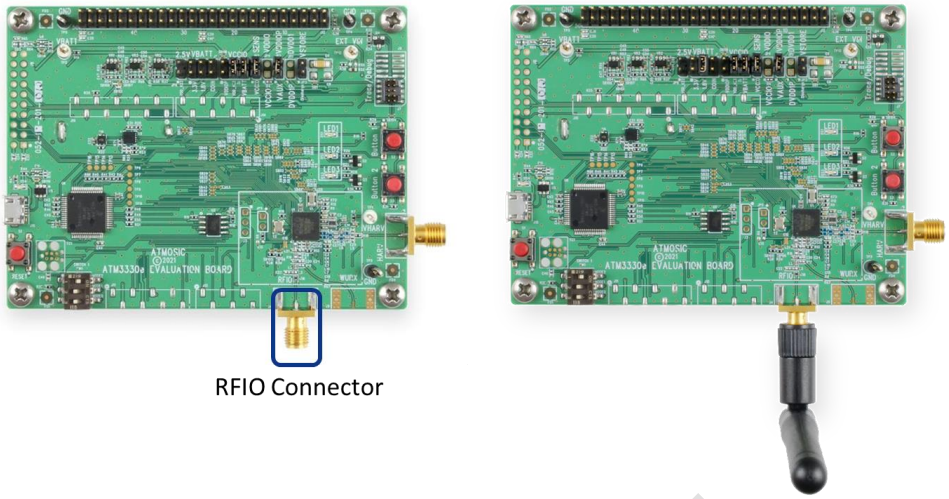


Figure 6 - 2.4 GHz Antenna Connection

Insert a coin cell battery into the battery holder on the backside of the board, make sure the '+' sign on the coin cell is facing upward. See Figure 7.

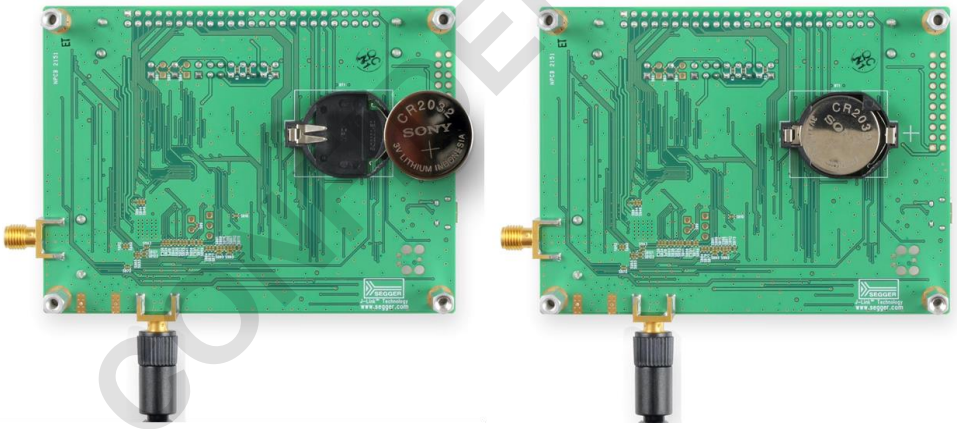


Figure 7 - Insert Coin Battery

After installing the coin cell, ensure that the power source jumper is in the COIN position to enable the tag.

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IMPORTANT NOTE:

As the tag is configured for low power, there is no visual LED indication that the board is transmitting, the GUI will validate this in a later step.

To conserve power when not in use, turn off the tag by either removing the coin cell or removing the jumper from the COIN position.

5. Set up the AoA Locator Board

Connect the locator board to the PC using the supplied USB cable.

Once connected, two LEDs will light:

- An LED adjacent to the USB connector on the ATM33/e evaluation board. This will indicate successful connection of the ATM33/e evaluation board to the PC.
- The left-most LED of the status LEDs on the antenna array board, which indicates that the antenna board is correctly powered by the ATM33/e board.

Return to the GUI and click ‘Try Again’ after the locator has been connected to the computer. See Figure 8.

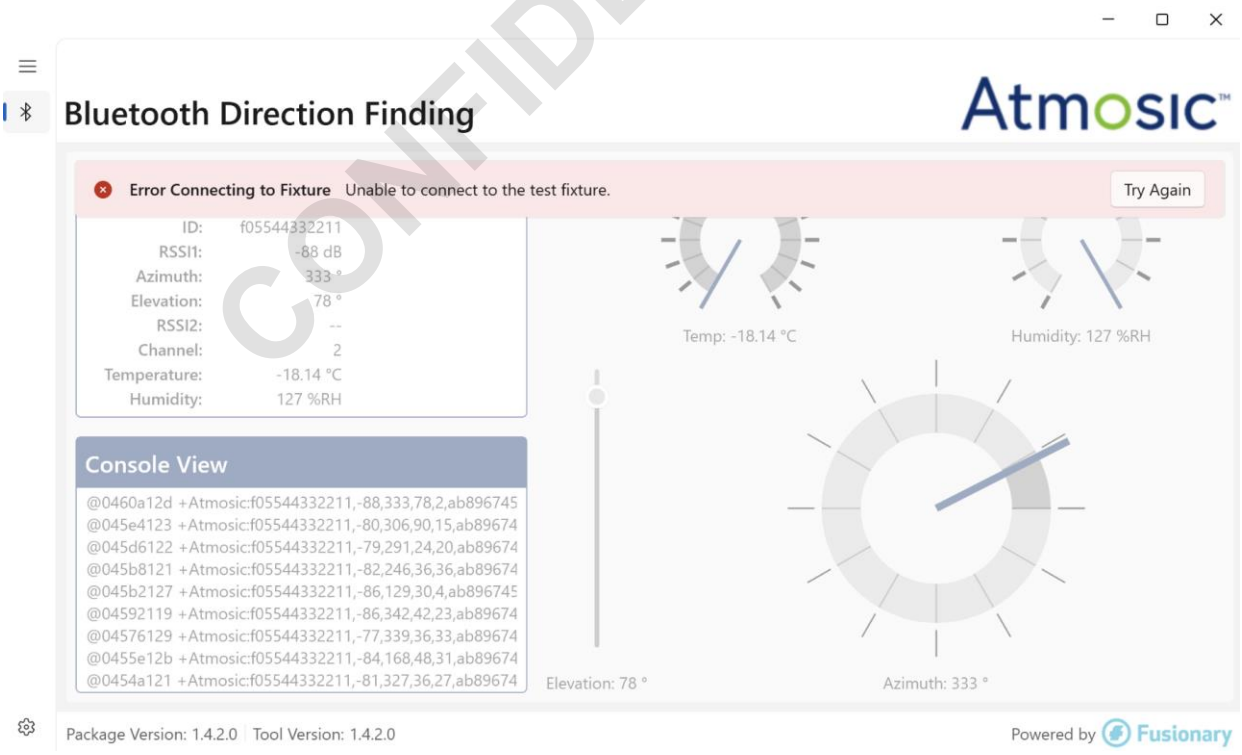


Figure 8 – ‘Try Again’ Button in the GUI Software

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Once the locator is success connected and the tag is powered, the GUI dashboard will become populated with data

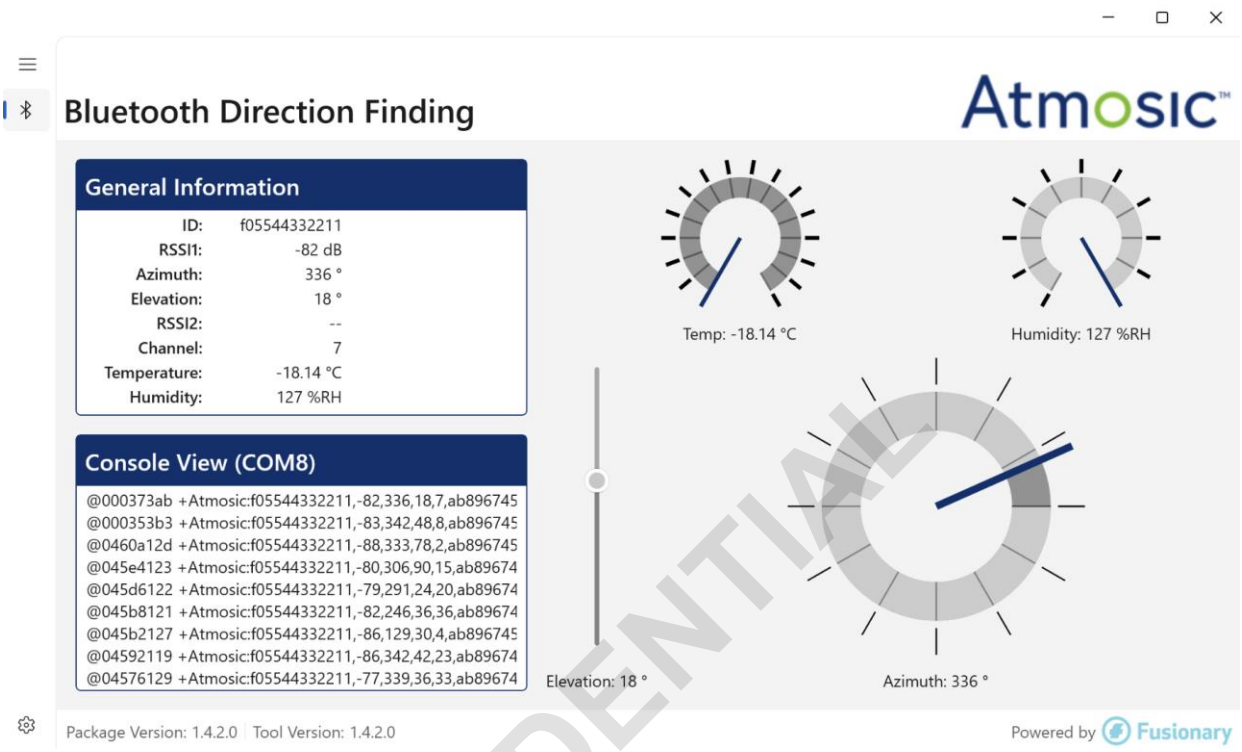


Figure 9 - GUI Dashboard

***Note**: The data displayed may vary slightly based on the firmware or hardware configurations of the tag and locator used. By example; In the figure shown above, temperature and humidity data is not supported by the particular tag/firmware combination and is therefore being reflected as 'no data'.*

6. Using the Demo

Position the tag next to the locator as shown in Figure 10. The position of the tag relative to the locator can be seen in the console view, general information, and the gauges.

The console view represents the data streaming into the PC via the USB cable which is decoded and highlighted both in the General Information section and on the gauges.

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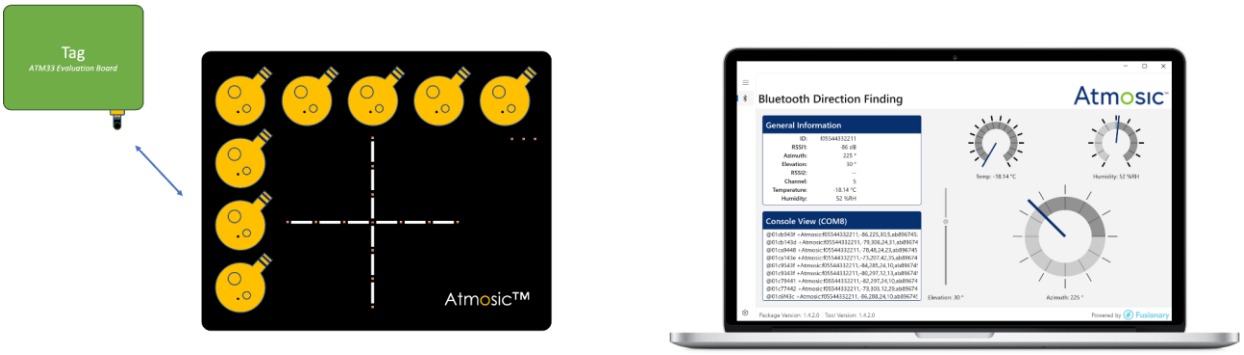


Figure 10 - GUI Display if Tag is Located on the left of the locator

Next, move the tag above the locator above the locator as shown in Figure 11. The data seen in both the general information and gauges sections will update to reflect the new tag position.



Figure 11 - GUI Display if Tag is Above the Locator

Figure 12 depicts the GUI display when the tag is moved to the right of the locator.

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Figure 12 - GUI Display if Tag is Located on the Right of the Locator

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7. Decoding the Data

The GUI displays the live location data of the tag. This includes the elevation of the tag (vertical angle relative to the locator) and the azimuth (horizontal angle relative to the locator).

In the General Information section of the GUI can be seen the ID, RSSI1, Azimuth, Elevation, RSSI2, Channel, Temperature, Humidity, Anchor ID, System Uptime, Device Type which is decoded from the raw data from the locator board seen in the Console View section. See Table 1 for description of the variables.

Table 1 - Variable Description

Variable	Description
ID	Identification code of the tag
RSSI1	RSSI value of the tag
Azimuth	Horizontal angle of tag relative to locator
Elevation	Vertical angle of tag relative to locator
RSSI2	Not displayed
Channel	Bluetooth channel
Temperature	Temperature of tag's environment
Humidity	Humidity of tag's environment
Anchor ID	Identification code of locator
System Uptime	Duration of time GUI has been active

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8. FAQ/Troubleshooting

8.1 The USB cable is connected but the GUI suggests there is no connection.

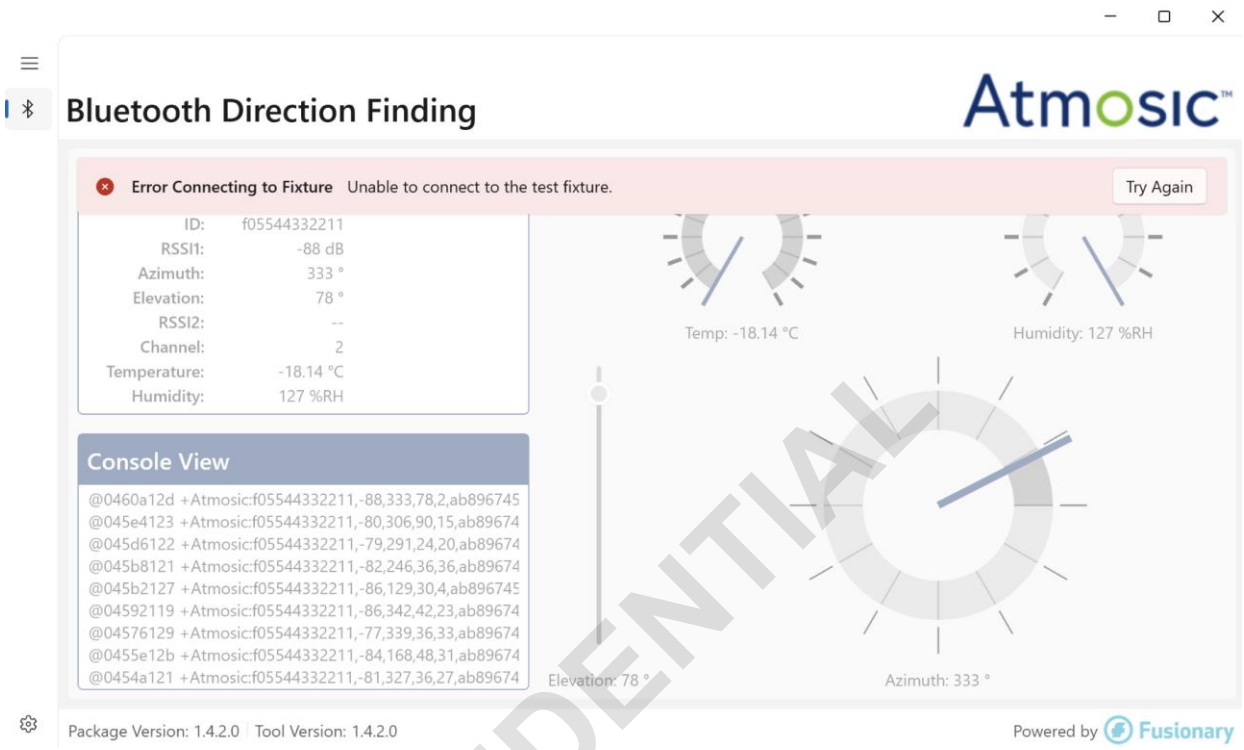


Figure 13 - USB Cable Connected But No Connection Recognised

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There are many reasons why the GUI might be getting the "**Error Connecting to Fixture Unable to connect to the test fixture**." message as shown in Figure 13.

1) Make sure that the locator is plugged in to the computer properly
2) Check to see if other debug tools such as PuTTY or Tera Term are using the COM ports disabling access to the GUI.

After addressing these potential issues, click on the "**Try Again**" button to see if the error is resolved.

8.2 The GUI is populated but doesn't appear to be updating.

Some data may have been initially received but then stopped if the tag is moved out of range or loses power. Verify that the tag is within range, the power jumper is in the COIN position and restart the GUI. If the issue persists, replace the battery.

8.3 Where can I find the source code for the GUI?

Please contact sales

8.4 Where can I find the example source code for the Tag?

The example source code is supplied in the SDK as of SDK version 5.2.

8.5 Where can I find the source code for the locator?

Please contact sales.

8.6 Why is there no data shown for temperature and humidity?

In support of several customer products, the GUI also supports several 3rd party tags and locators in addition to the Atmosic evaluation design. The configuration of these 3rd party products may not support these additional features. Be sure to use Atmosic reference hardware and firmware to see functionality as described in this documentation.

8.7 Are the supplied ATM33/e (052-311-xxx) evaluation boards standard?

The EVB's supplied in this kit has been pre-loaded with application firmware and jumpers are pre-configured for the AoA demo.

Otherwise, the hardware is standard.

8.8 The display does not align with the evaluation board position, why?

Remember that the signal comes from the antenna, not the evaluation board, as such the locator will be pointing to the antenna not the board itself.

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Revision History

Date	Version	Description
September 6, 2022	0.60	Initial version created.

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